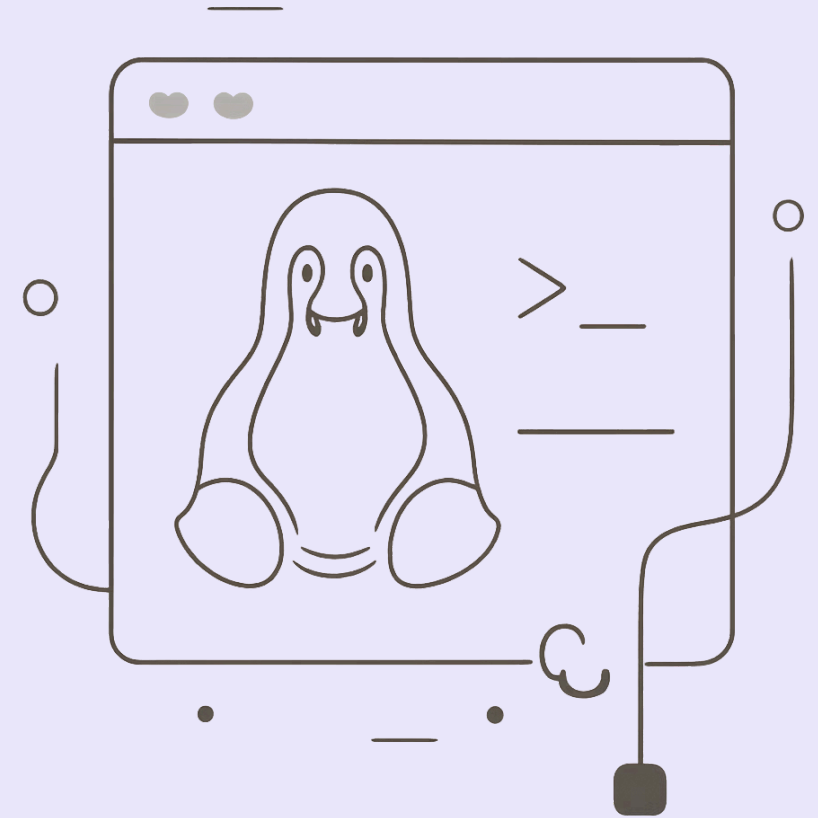


OPERATING SYSTEMS & SYSTEMS PROGRAMMING | 25CS2104E

Linux System Information and Resource Monitoring Tool

SECTION: 1 | KERNEL_KINGS_OSSP





Team Members

V. Navadeep

Reg. No: 2520030019

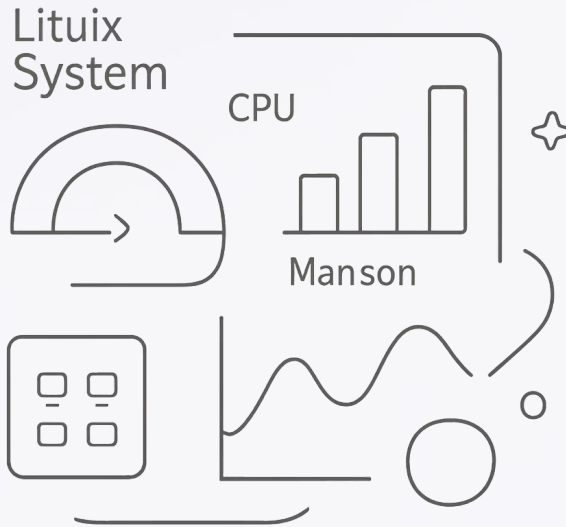
K. Shankar

Reg. No: 2520030063

CH. Sai Hima Sekhar

Reg. No: 2520030554

Abstract



The **Linux System Information and Resource Monitoring Tool** is a system-level application that collects and displays real-time data about a Linux system's health and performance. A Linux system is continuously managing different resources such as the CPU, memory, disk, and running processes. Our tool focuses on monitoring these resources and presenting their status to the user. The important part of our project is that instead of depending only on external monitoring applications, our program directly interacts with Linux system interfaces, particularly the `/proc` filesystem, to obtain system information.

→ Monitors

CPU usage, memory, disk usage, and running processes

→ Interface

Reads directly from Linux system interfaces like `/proc`

→ Output

Presents data in a simple, user-friendly terminal format



Problem Statement

Linux systems generate vast amounts of system and resource information, but accessing and interpreting this data manually is challenging — especially for beginners.

Fragmented Information

Users must run multiple commands to gather CPU, memory, and process details separately.

Raw, Unformatted Output

System data is presented as raw text, making it difficult to interpret quickly.

Overly Complex Tools

Existing monitoring tools often provide excessive information beyond basic system analysis needs.



Proposed Methodology

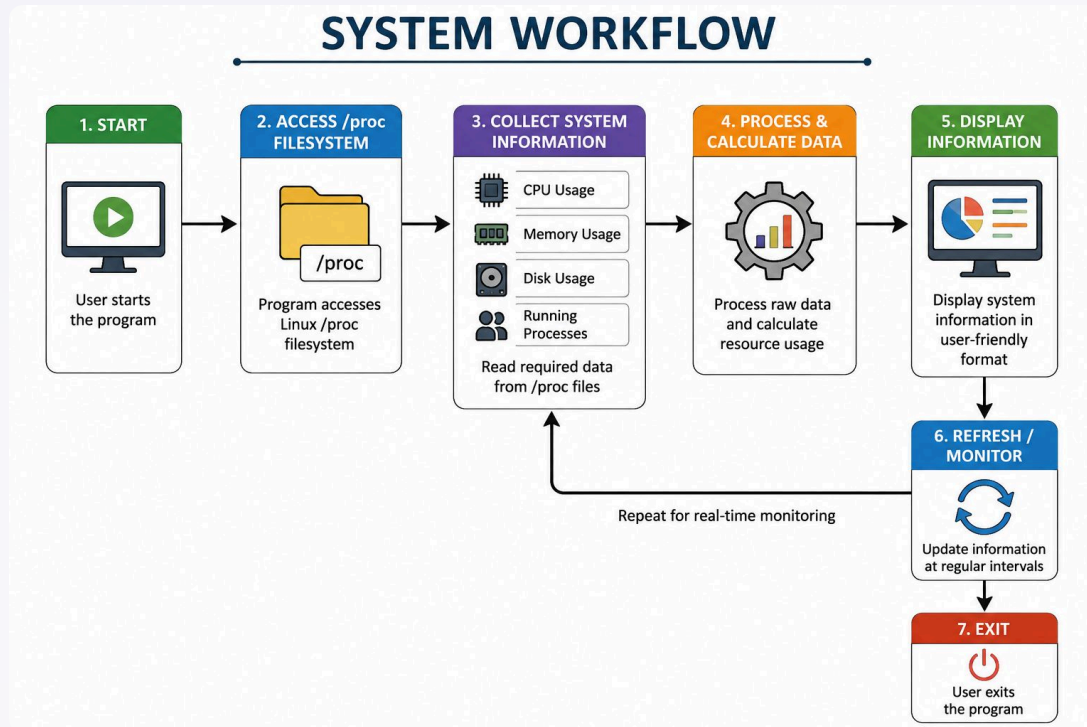
The project follows a **Linux system-level monitoring approach** using the `/proc` filesystem.

The tool is developed in **C** to read and process system information and present it in a simple terminal-based format.

1. 📁 **Access `/proc` Filesystem** – Read kernel-provided system information.
2. 📊 **Collect Resource Data** – Obtain CPU, memory, disk, and process information.
3. ⚙️ **Process & Calculate Data** – Analyze the collected values to determine resource usage.
4. 🖥️ **Display Results** – Present the processed information in a clear, user-friendly format.
5. 🔄 **Monitor Continuously** – Refresh the data to provide updated resource information.

Workflow Steps

- ▶ Start the Application
- 📁 Access Linux /proc Filesystem
- 📊 Collect System Information
 - CPU Usage
 - Memory Usage
 - Disk Usage
 - Running Processes
- ⚙️ Process & Calculate Resource Data
- 📺 Display Information to User
- 🔄 Refresh / Monitor Updated Information
- 🔴 Exit Application



Tools and Technologies Used



Operating System

Linux / Ubuntu — the target platform for development and deployment



Programming Language

C — low-level access to system calls and file interfaces



System Interface

/proc filesystem — virtual filesystem exposing kernel and process data



Terminal & Compiler

Linux Terminal for execution; GCC for compiling C source code



Code Editor

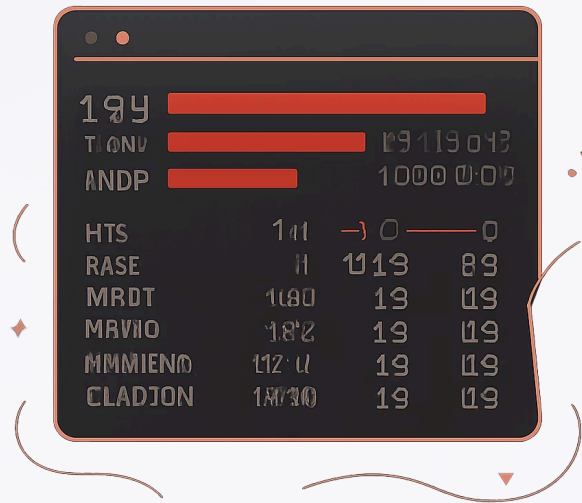
VS Code / Vim — used for writing and debugging the C source



System Concepts

Process Management, CPU & Memory Monitoring, File System Management

Expected outcomes.



The project is expected to provide a **simple and efficient tool** for monitoring Linux system resources. It will collect real-time system information using the `/proc` **filesystem** and present it in an easy-to-understand format. The tool will help users observe **CPU, memory, disk, and process usage**. It also provides practical understanding of **Linux system programming concepts**.

1  **Real-time system resource monitoring**

2  **Simple and user-friendly information display**

3  **Practical understanding of Linux system programming**

Thank You

We appreciate your time and attention. Questions and feedback are welcome.

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